Lesson 1 — Extra Practice

Graphing and Classifying Systems of Linear Equations

Part A — Mixed Forms: Rearrange Before You Graph

*In every system below one equation already sits in slope y-intercept form, $y = mx + b$, and the other is in general form, $Ax + By + C = 0$. General form hides the slope and the y-intercept, so **nothing can be graphed until the second equation is rearranged**. Do the rearranging in question 1; do the graphing in question 2.*

1. Rewrite the general-form equation of each system in slope y-intercept form, then state the slope and the y-intercept of *both* lines of the system.

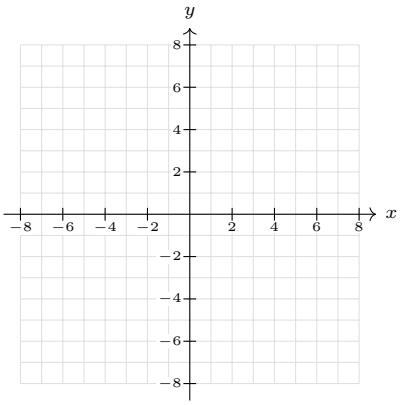
(a) $y = -\frac{3}{2}x + 4$
$3x - 4y - 20 = 0$

(b) $y = \frac{1}{3}x - 1$
$2x + 3y - 15 = 0$

(c) $y = -x + 2$
$5x - 2y + 18 = 0$

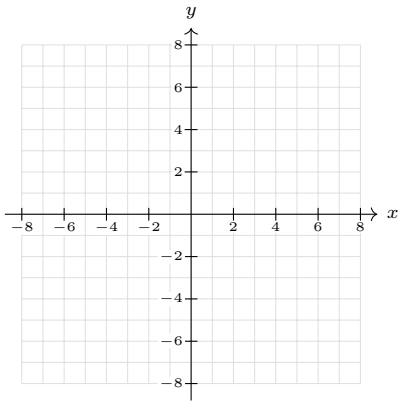
2. Graph both lines of each system from question 1 on the matching grid below. State the solution as an ordered pair, and verify it by substituting into **both original** equations.

(a)



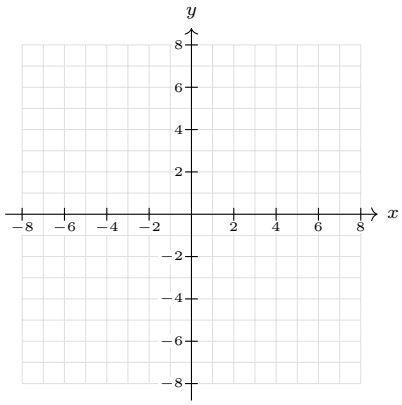
(a) Solution and verification:

(b)



(b) Solution and verification:

(c)



(c) Solution and verification:

Part B — Classifying a System Without Graphing

*A system of two linear equations has **one solution** when the slopes differ, **no solution** when the slopes are equal and the y-intercepts differ, and **infinitely many solutions** when the slopes and the y-intercepts are both equal. For a line written in general form $Ax + By + C = 0$ the slope is $-\frac{A}{B}$ and the y-intercept is $-\frac{C}{B}$, so both numbers can be read straight out of the coefficients — no graph, and no full rearrangement, required.*

3. For each system, use $-\dfrac{A}{B}$ and $-\dfrac{C}{B}$ to state the slope and y-intercept of each line, then state whether the system has one solution, no solution, or infinitely many solutions. Say *why* in each case.

(a) $2x - 5y + 10 = 0$
$4x - 10y - 15 = 0$

(b) $3x + 4y - 12 = 0$
$9x + 12y - 36 = 0$

(c) $x - 6y + 7 = 0$
$6x - y + 7 = 0$

(d) $5x + 2y - 8 = 0$
$15x + 6y - 24 = 0$

(e) $7x - 3y + 21 = 0$
$14x - 6y + 21 = 0$

(f) $4x + 9y - 18 = 0$
$9x + 4y - 18 = 0$

4. Four lines are given in general form.

$$\ell_1 : \ 6x - 15y + 9 = 0 \qquad \ell_2 : \ 2x - 5y + 3 = 0 \qquad \ell_3 : \ 6x - 15y + 3 = 0 \qquad \ell_4 : \ 2x - 3y + 9 = 0$$

(a) Two of these four equations look quite different but describe the **same line**. Identify them, and state the multiplier that turns one into the other.

(b) ℓ_1 and ℓ_3 differ in one digit only. Classify the system formed by ℓ_1 and ℓ_3, and justify your answer using $-\dfrac{A}{B}$ and $-\dfrac{C}{B}$.

(c) ℓ_2 and ℓ_4 also look almost alike. Classify the system formed by ℓ_2 and ℓ_4, and, if it has one, find the solution and verify it in both equations.

(d) In one or two sentences, explain what you must check — and in what order — to tell a pair of equations that is *one line written twice* from a pair that is *two parallel lines*.

Part C — Solving for an Unknown Coefficient

*"No solution" asks for **one** condition: matching slopes. "Infinitely many solutions" asks for **two**: matching slopes* and *matching constants — every coefficient must scale by the same multiplier. Watch what happens when the second condition cannot be met.*

5. Consider the system $kx + 4y - 12 = 0$ and $6x + 8y - 40 = 0$.

(a) Determine the value of k for which the system has **no solution**. Confirm your value by writing both lines in $y = mx + b$ form.

(b) Show that **no** value of k makes the system have infinitely many solutions. Name the condition that fails.

(c) State the values of k for which the system has exactly one solution.

6. Consider the system $2x - 5y + 8 = 0$ and $6x + ky + 24 = 0$.

(a) Determine the value of k for which the system has **infinitely many solutions**, and state the multiplier that links the two equations.

(b) Explain why that same value of k cannot give no solution, referring to the constant terms.

(c) Replace the constant 24 by a number that makes the system have **no solution** when $k = -15$, and justify your choice.

7. Consider the system $6x + ky + 21 = 0$ and $2x - 3y + c = 0$, where k and c are constants.

(a) Determine the values of k and c for which the system has infinitely many solutions.

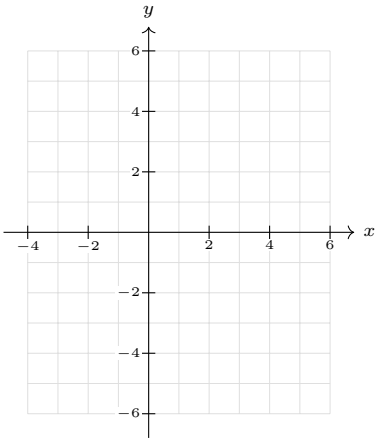
(b) State the values of k and c for which the system has no solution, and the values for which it has exactly one solution.

Part D — When the Intersection Misses the Lattice Points

*A graph can only be read as accurately as it is drawn. When the lines cross between the grid lines, the best a graph can give is an **estimate** — and an honest answer says so. This is exactly why Lessons 2 and 3 replace the graph with algebra.*

8. Consider the system $y = -\frac{1}{2}x + 3$ and $3x - 2y - 4 = 0$.

(a) Rearrange the second equation and graph both lines.



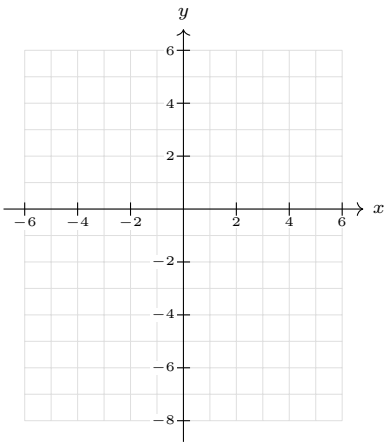
(b) Estimate the solution from your graph, reading each coordinate to the nearest half unit. Then explain, in one sentence, why your answer is an estimate and not the solution.

(c) Substitute your estimate into both original equations. Show that at least one of them fails to balance.

(d) The exact solution is $\left(\frac{5}{2}, \frac{7}{4}\right)$. Verify it in both original equations, and state how far your graphical estimate was from it.

9. Consider the system $y = \frac{2}{3}x - 1$ and $4x + 3y + 9 = 0$.

(a) Rearrange the second equation and graph both lines.



(b) One coordinate of the solution can be read exactly off the grid and the other cannot. State which is which, and give your best estimate of the solution.

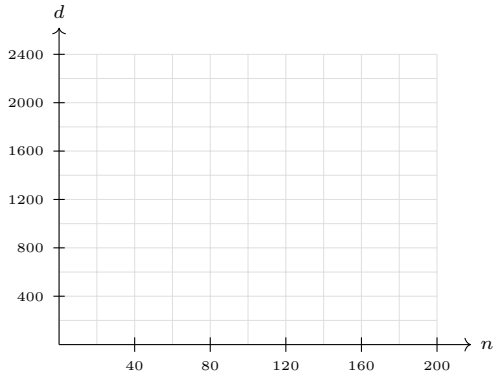
(c) Explain what feature of the two slopes makes the x-coordinate land on a grid line while the y-coordinate does not.

(d) Show that $\left(-1, -\frac{5}{3}\right)$ satisfies both original equations, and state the solution of the system as an ordered pair.

Part E — A Break-Even Point Read From a Graph

10. A student council orders reusable water bottles printed with the school crest. The order costs a fixed \$800 set-up charge plus \$4 for each bottle printed, and the council sells the bottles for \$12 each. Let n be the number of bottles and d the amount in dollars.

(a) Write the cost equation and the revenue equation, then graph both on the grid.



(b) Read the break-even point off the graph and interpret it in a complete sentence, with units.

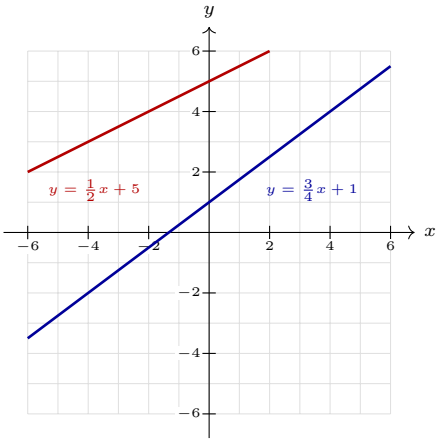
(c) Determine the profit or loss made on 60 bottles and on 175 bottles. Answer each in a sentence with units.

(d) Without graphing anything further, explain how the two slopes guarantee that this system has exactly one break-even point.

Part F — Error Analysis

11. Asked how many solutions the system $y = \frac{3}{4}x + 1$ and $y = \frac{1}{2}x + 5$ has, a student graphs both lines by hand on a grid with $-6 \le x \le 6$ and $-6 \le y \le 6$, sees no crossing point, and writes *"the lines do not meet, so the system has no solution."*

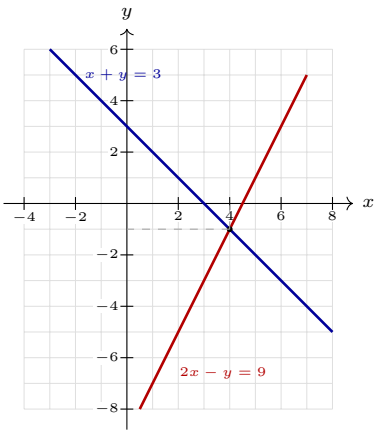
(a) Classify the system correctly *without* graphing, and state in one sentence what the student's error was.



(b) Determine the point of intersection algebraically, and verify it in both equations.

(c) State a grid — a range of x values and a range of y values — on which the intersection would be visible, and explain why an empty grid can never by itself prove that a system has no solution.

12. The system $x+y=3$ and $2x-y=9$ has been graphed below, and the point of intersection is marked. A student reports the solution as $(-1,4)$.



(a) Substitute $(-1,4)$ into both equations. Show that it satisfies one of them but not the other, and name the mistake the student made in reading the graph.

(b) State the correct solution as an ordered pair and verify it in both original equations.

(c) The student's pair passed one of the two checks. Explain why checking in **both** equations — not just one — is what catches this error.

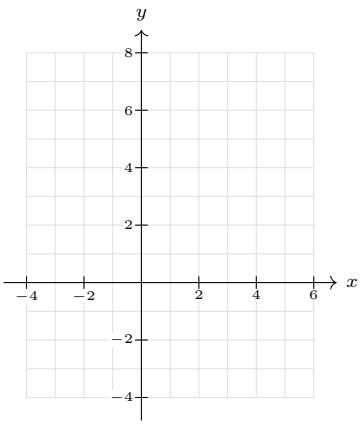
Part G — Stretch: Three Lines at Once

*Three lines are said to be **concurrent** when all three pass through a single common point. With three lines there are three pairs to consider, and each pair is a system of its own.*

13. Three lines are given in general form.

$$\ell_1 :\ x - y - 1 = 0 \qquad \ell_2 :\ 2x + y - 8 = 0 \qquad \ell_3 :\ 3x - 3y + 6 = 0$$

(a) Write each line in slope y-intercept form and draw all three on the grid, labelling them.



(b) Exactly two of the three pairs of lines intersect. State which pair does *not*, and classify that pair's system.

(c) Determine the point of intersection of each of the two pairs that do meet, and verify each point in both of its equations.

(d) Are the three lines concurrent? Answer in a sentence, and justify it from your answers to (b) and (c).

(e) Determine the equation, in general form, of a line ℓ_4 that is not parallel to ℓ_1 or ℓ_2 and that makes ℓ_1, ℓ_2 and ℓ_4 concurrent. Verify that your line works.

Part H — Multiple Choice and Numerical Response

14. *(Multiple Choice)* The system $\begin{cases} 6x - 4y + 14 = 0 \\ 9x - 6y + 21 = 0 \end{cases}$ has

A. no solution B. exactly one solution C. exactly two solutions D. infinitely many solutions

15. *(Multiple Choice)* The value of k for which the lines $kx + 12y - 5 = 0$ and $3x - 4y + 9 = 0$ have no point in common is

A. -16 B. -9 C. 9 D. 16

16. *(Numerical Response)* The lines $y = \frac{5}{2}x - 7$ and $4x + 3y - 25 = 0$ intersect at the point (p, q). The value of $p + q$ is _____. *(Record your answer as a three-digit number.)*

17. *(Numerical Response)* The system $8x + ky - 20 = 0$ and $2x + 5y - 6 = 0$ has no solution when $k =$ _____. *(Record your answer as a three-digit number.)*

Lesson 2 — Extra Practice

Solving Systems of Equations by Substitution

Part A — Choosing What to Isolate

Substitution always works, but it does not always work pleasantly. *A coefficient of* 1 *or* -1 *isolates with no division at all; every other coefficient divides and leaves a fraction behind. Before you solve anything in question 1, look at all four coefficients and pick the one that costs you nothing.*

1. For each system: **(i)** name the equation and the variable you chose to isolate and say in a few words *why*; **(ii)** solve by substitution; **(iii)** check the solution in **both** original equations.

(a) $5x + y = 11$
$3x - 4y = 2$

(b) $x - 3y = 2$
$4x + 5y = 42$

(c) $6x - y = 13$
$4x + 3y = 5$

(d) $9x + 2y = 5$
$x + 4y = 27$

(e) In part (c) the coefficient you wanted was -1, not 1. Explain why -1 is just as good a choice as 1, and what the one extra thing is that you must watch when you isolate from it.

2. In each system below **no** coefficient is 1 or -1, so a fraction is unavoidable. Solve each one by substitution, **carrying the fractions exactly** — no rounding and no decimals at any stage — and check in both original equations.

(a) $3x + 2y = 8$
$5x - 4y = 6$

(b) $6x - 4y = 9$
$4x + 2y = 5$

(c) $4x + 6y = 7$
$2x - 5y = 9$

(d) Part (a) had a fractional *middle* but an integer *answer*. Explain how that is possible, and why a fraction appearing partway through a solution is never by itself a sign that you have made an error.

Part B — Both Equations in General Form

General form, $Ax + By + C = 0$, hides every variable inside the equation. Nothing can be substituted until something has been isolated, so isolating is the first *step here, not an afterthought.*

3. Solve each system by substitution. State which variable you isolated and from which equation, then check the solution in both original equations. Write each answer as an ordered pair.

(a) $3x - y - 7 = 0$
$5x + 2y - 19 = 0$

(b) $4x + 3y + 6 = 0$
$x - 2y - 4 = 0$

(c) $2x + 5y + 4 = 0$
$3x - 4y - 17 = 0$

(d) $6x - 5y - 3 = 0$
$2x + 3y + 13 = 0$

Part C — Clear the Fractions and Decimals First

A fraction or a decimal in a coefficient is not a reason to reach for a calculator. Multiply the whole equation by a number that clears it — the lowest common denominator for fractions, a power of 10 *for decimals — and you are back to integers before you begin.*

4. Clear the fractions or decimals from each equation first, then solve by substitution. Check each solution in the **original** (uncleared) equations.

(a) $\frac{x}{2} - \frac{y}{4} = 2$
$\frac{x}{3} + \frac{y}{2} = 4$

(b) $0.4x + 0.3y = 1.4$
$0.2x - 0.5y = 2$

(c) $0.5x - 0.25y = 1.5$
$\frac{x}{3} + \frac{2y}{3} = 2$

(d) A classmate says part (b) would be easier with a calculator, working in decimals throughout. Give one reason the cleared-integer version is safer, and say what that calculator habit would have done to question 2(b).

Part D — When Substitution Collapses

Sometimes every variable disappears at once. What is left is either a statement that is always true, such as $0 = 0$, or one that is never true, such as $0 = 7$. Those two endings are not failures — each one is the answer, and each one says something exact about the two lines.

5. Solve each system by substitution. Each one collapses. For each: **(i)** write the final collapsed statement, **(ii)** state whether the system has **one solution**, **no solution**, or **infinitely many solutions**, and **(iii)** say what that means about the two lines.

(a) $y = 3x - 4$
$9x - 3y = 12$

(b) $3x - 7y = 5$
$6x - 14y = 4$

(c) $x + 4y = 7$
$3x + 12y = 21$

(d) $3x + 2y - 5 = 0$
$6x + 4y + 1 = 0$

(e) A student says "I got $x = 0$, so there is no solution." Explain, in one or two sentences, the difference between finding $x = 0$ and arriving at $0 = 7$, and what each one tells you.

Part E — Answers That Are Not Lattice Points

Lesson 1 found solutions by graphing. A graph can only ever report a point it can land on; substitution is not limited that way. The two systems below have solutions no grid would have given you.

6. Solve each system by substitution, leaving every coordinate as an exact fraction in lowest terms. Verify each solution in **both** original equations, working with fractions throughout.

(a) $3x + 2y = 4$
$5x - 3y = 1$

(b) $7x - 3y = 5$
$2x + 5y = 4$

(c) Both solutions sit between 0 and 1 in each coordinate. Explain what a graph drawn with one square per unit would have reported for part (b), and why substitution is the method that has to be trusted here.

Part F — Translating to a System

The mathematics in questions 7 and 8 is the mathematics you have already done. The work is in naming the two unknowns, writing the two sentences the problem gives you as two equations, and finishing with an answer a reader could act on.

7. A lab technician has one bottle of 25% acid solution and one bottle of 60% acid solution. She needs to mix them to make 700 mL of a 40% acid solution.

(a) Define the two variables, stating the units, and write a system of two equations — one for volume and one for acid.

(b) Say which variable you will isolate and why, then solve the system by substitution.

(c) Check both equations, and state the answer as a sentence with units.

8. A river tour boat travels 48 km downstream in 2 hours, then returns the same 48 km upstream in 3 hours. Downstream the current adds to the boat's own speed; upstream it subtracts.

(a) Let b be the boat's speed in still water and c the speed of the current, both in km/h. Write two equations using distance = speed $\times$ time.

(b) Solve the system by substitution, and check both equations.

(c) State both answers as a sentence with units, and say how long the trip would take with no current at all.

Part G — Error Analysis

9. Asked to solve $2x + y = 7$ and $3x - 4y = 5$, a student writes

$$2x + y = 7 \;\Rightarrow\; y = 7 - 2x, \qquad \text{then} \qquad 2x + (7 - 2x) = 7 \;\Rightarrow\; 7 = 7,$$

and concludes that the system has infinitely many solutions.

(a) Name the error in one sentence, and explain why $7 = 7$ was guaranteed before the student even started.

(b) Carry out the substitution correctly and solve the system, checking in both equations.

(c) The student's conclusion can also be refuted by a single counter-example. Find a point that satisfies $2x + y = 7$ but not $3x - 4y = 5$, and say what it proves.

10. Asked to solve $5x - 2y = 4$ and $x + 3y = 11$, a student isolates $y = \dfrac{11 - x}{3}$, substitutes correctly, reaches $17x = 34$, writes "$x = 2$", and stops.

(a) Reproduce the student's work as far as $x = 2$, showing how the fraction is cleared.

(b) Finish the question properly and check in both equations.

(c) Explain why "$x = 2$" cannot be the solution of a system of two equations in two variables, and what a solution must look like.

11. Asked to solve $x + 2y = 3$ and $4y - x = 15$, a student isolates $x = 3 - 2y$ and then writes

$$4y - 3 - 2y = 15 \;\Rightarrow\; 2y = 18 \;\Rightarrow\; y = 9, \qquad x = 3 - 2(9) = -15.$$

(a) Write the substitution the way it should have been written, with the bracket in place, and say exactly which sign the missing bracket changed.

(b) Solve the system correctly and check in both equations.

(c) Test the student's point $(-15, 9)$ in each original equation. It passes one and fails the other — explain why that is exactly what a dropped bracket in the *second* equation should produce.

Part H — Stretch

12. Consider the system

$$\frac{x}{3} - \frac{y}{2} = \frac{5}{6} \qquad\qquad 4x - 6y - 10 = 0.$$

(a) Clear the fractions from the first equation and reduce the second to lowest terms. What do you notice?

(b) Solve by substitution anyway, and state the collapsed statement, the classification, and what it says about the lines.

(c) Give three different ordered pairs that satisfy **both** equations, and describe the complete solution set in words.

13. In the system below, k is a constant, not a variable.

$$x + ky = 3 \qquad 2x - y = 1$$

(a) Which equation should be rearranged, and for which variable? Solve the system by substitution to express x and y in terms of k.

(b) Verify your expressions in **both** original equations, showing that every k cancels.

(c) Every answer in (a) has the same denominator. For what value of k does that denominator become 0? Substitute that k into the original system and classify it.

(d) Use your formulas to write down the solution when $k = 2$, then check that ordered pair in both equations of the original system with $k = 2$ substituted in.

Part I — Multiple Choice and Numerical Response

14. *(Multiple Choice)* A student is about to solve the system $3x - 7y = 12$ and $5x + y = 8$ by substitution. The *only* one of the following rearrangements that introduces no fraction is

A. $x = \dfrac{12 + 7y}{3}$ B. $y = 8 - 5x$ C. $y = \dfrac{3x - 12}{7}$ D. $x = \dfrac{8 - y}{5}$

15. *(Multiple Choice)* The system $4x - 6y = 10$ and $6x - 9y = c$ has infinitely many solutions when c equals

A. 10 B. 12 C. 15 D. 20

16. *(Numerical Response)* The system $7x + 2y = 40$ and $x - 3y = -14$ is solved by substitution. The value of $12x + 5y$ is _____. *(Record your answer as a three-digit number.)*

Lesson 3 — Extra Practice

Combining Equations: The Elimination Method

Part A — Eliminating Directly: Add, or Subtract?

*Every system in Part A is already lined up: one variable has coefficients that are either **equal** or **opposite**, so Step 1 costs nothing. The only decision is **add or subtract**, and it is decided by the coefficients, not by the signs in front of the constants. Write the combination as a bracketed difference, $(\cdots) - (\cdots)$, whenever you subtract.*

1. For each system, name the variable that disappears, say whether you must **add** or **subtract** and why, then solve. Verify every solution in **both** original equations.

(a) $\begin{aligned} 7x + 4y &= 16 \\ 5x - 4y &= 32 \end{aligned}$ (b) $\begin{aligned} 6x - 5y &= -37 \\ 6x + 3y &= 3 \end{aligned}$ (c) $\begin{aligned} -9a + 2b &= -35 \\ 9a + 7b &= -1 \end{aligned}$ (d) $\begin{aligned} 11m - 6n &= 39 \\ -4m - 6n &= -6 \end{aligned}$

2. The two systems below are identical except for **one sign**: the $5y$ in the second equation. Solve each one.

(a) $\begin{aligned} 3x + 5y &= 16 \\ 7x + 5y &= 4 \end{aligned}$ (b) $\begin{aligned} 3x + 5y &= 16 \\ 7x - 5y &= 4 \end{aligned}$

(c) State, in one sentence, the rule that decides whether to add or to subtract. Then explain why changing that single sign moved the solution as far as it did.

Part B — Scaling One Equation

*If no variable is ready as the system stands, look for a coefficient that divides into its partner. Multiplying **one** equation by a whole number is then enough. Remember that multiplying an equation means multiplying* every *term — the constant included.*

3. Solve each system by scaling **one** equation. State the multiplier you used and which equation you applied it to, then verify in both originals.

(a) $\begin{aligned} x - 4y &= 13 \\ 3x + 5y &= 5 \end{aligned}$ (b) $\begin{aligned} 2m + 9n &= -5 \\ 6m - 5n &= 17 \end{aligned}$ (c) $\begin{aligned} 7u + 3v &= 1 \\ u - 6v &= 13 \end{aligned}$

Part C — Scaling Both Equations

*When neither coefficient divides into the other, scale both. The smallest pair of multipliers comes from the **lowest common multiple** of the two coefficients: divide the LCM by each coefficient in turn to get the two multipliers.*

4. Solve each system by scaling **both** equations. State the LCM you used and the two multipliers, then verify in both originals.

(a) $4x + 7y = -2$
$6x - 5y = 28$

(b) $9p + 4q = 7$
$6p - 11q = -50$

(c) $10x + 3y = 5$
$4x - 7y = 43$

Part D — Which Variable Is Cheaper to Eliminate?

Both variables can always be eliminated, but one of them is usually far less work. Count the multiplications before you start.

5. Consider the system $3x + 8y = 4$ $7x + 4y = 24$.

(a) Write the multipliers needed to eliminate x, and then those needed to eliminate y. Which variable is cheaper, and why?

(b) Eliminate the cheaper variable and solve the system. Verify in both original equations.

(c) Go back to the original system and eliminate the *other* variable instead. Confirm you reach the same ordered pair.

6. For each system, state which variable is cheaper to eliminate and give the *exact* multipliers each of the two choices would need. Do not solve in (a)–(c).

(a) $9a - 4b = 30$
$3a + 5b = -9$

(b) $6x - 5y = 9$
$4x + 15y = 61$

(c) $5m + 8n = 2$
$5m - 3n = 13$

(d) Now solve system (b) by the cheaper route, and verify in both originals.

Part E — Rearranging Into $Ax + By = C$ First

Nothing lines up until the variables are on the same side, in the same order, in both equations. Rearrange first; only then compare coefficients.

7. Rewrite each equation in the form $Ax + By = C$, then solve by elimination and verify in both *original* (unrearranged) equations.

(a) $5x - 3y = 11$
$2x = -7y - 12$

(b) $3(x + 2y) = 15$
$2x + 22 = 4y$

(c) $6a - 5b - 15 = 0$
$4b = 2a + 2$

Part F — Clearing Fractions and Decimals First

Multiply each equation through by whatever makes every coefficient an integer — the lowest common denominator, or a power of ten — before *you compare coefficients. Two separate equations may need two different multipliers.*

8. Clear the fractions or decimals, solve by elimination, and verify your solution in the *original* equations as given.

(a) $\frac{x}{4} - \frac{y}{3} = \frac{1}{2}$
$\frac{x}{6} + \frac{y}{2} = \frac{5}{2}$

(b) $0.3x - 0.4y = 1.7$
$0.5x + 0.2y = 1.1$

(c) $\frac{2x - y}{3} + \frac{x + y}{4} = \frac{7}{2}$
$\frac{x}{2} - \frac{3y}{4} = \frac{1}{2}$

Part G — When Both Variables Vanish

Sometimes the elimination step removes both *variables at once and leaves a bare numerical statement. Whether that statement is true or false is the whole answer.*

9. Solve each system by elimination. State whether it has **one solution**, **no solution**, or **infinitely many solutions**, and say why in terms of the two lines.

(a) $6x - 15y = 9$
$-4x + 10y = 7$

(b) $2x - 6y = 8$
$-3x + 9y = -12$

(c) $9x - 6y = 3$
$6x - 5y = 4$

(d) In two of these, both variables vanished. Explain how the surviving statement alone tells you which case you are in.

Part H — Elimination or Substitution?

Both methods are exact and both always work. Speed is the only thing that separates them, and you can judge it by looking rather than by solving.

10. For each system below, state which method — elimination or substitution — you would use, and give one sentence of reason. **Do not solve them here.**

(a) $y = 4x - 9$
$3x + 2y = 15$

(b) $7x + 5y = 41$
$7x - 3y = 9$

(c) $x - 3y = 7$
$4x + 5y = 11$

(d) $6x + 5y = -8$
$9x - 4y = 34$

11. Now solve systems **10(b)** and **10(d)** by the method you chose for each, and verify both solutions in both original equations.

(a) system 10(b)

(b) system 10(d)

Part I — Error Analysis

12. Asked to solve $5x + 4y = 21$ and $x - 2y = 7$ by elimination, a student wrote

$$\text{Eq. } 2 \times 2: \quad 2x - 4y = 7 \qquad \text{Eq. } 1 + \text{Eq. } 2: \quad 7x = 28,\ x = 4 \qquad 5(4) + 4y = 21,\ y = \tfrac{1}{4}.$$

(a) Name the error in one sentence, and write Eq. 2 scaled correctly.

(b) Solve the system correctly and verify in both original equations.

(c) Substitute the student's $\left(4, \frac{1}{4}\right)$ into *both* originals. Explain why checking in only one equation could never have caught this mistake.

13. Asked to solve $3x + 5y = 4$ and $2x - 5y = 11$, a student wrote: "Both equations contain $5y$, so I subtract. Eq. 1 − Eq. 2 gives $x = -7$."

(a) Carry out Eq. 1 − Eq. 2 correctly. Is y eliminated? State the rule that decides add or subtract, and which case this system is.

(b) Solve the system correctly and verify in both original equations.

(c) The student's equation is a *true* consequence of the system. Show that the correct solution satisfies it, then explain why a true combination can still be useless.

14. Asked to solve $2x + 5y = 16$ and $6x - 7y = 4$, a student wrote

$$\text{Eq. } 1 \times 3: \quad 6x + 15y = 48 \qquad (\text{Eq. } 1 \times 3) - \text{Eq. } 2: \quad 22y = 44,\ y = 2 \qquad 6x + 15(2) = 16,\ x = -\tfrac{7}{3}.$$

(a) Verify that the student's value $y = 2$ is correct.

(b) Say exactly what went wrong at the back-substitution step, name the two equations that are always safe to substitute into, and find x correctly.

(c) Show that the student's pair $\left(-\frac{7}{3}, 2\right)$ fails *both* originals, then verify your own solution in both.

Part J — Stretch

15. Determine the values of A and B for which the ordered pair $(3, -2)$ is the solution of the system

$$Ax + By = 14 \qquad Bx - Ay = 5$$

(a) Substitute $(3, -2)$ into each equation to form a system of two equations in A and B.

(b) Solve that system by elimination.

(c) Write out the original system with your values of A and B, and verify that $(3, -2)$ satisfies both equations.

16. In the system $kx + 6y = 9$ $4x + 3y = 7$, k is a constant.

(a) Eliminate y and show that the result is $(k - 8)x = -5$.

(b) For what value of k does the system have **no solution**? Justify it twice: from the surviving statement, and from the slopes.

(c) Is there a value of k giving **infinitely many solutions**? Justify your answer.

(d) Take $k = 3$ and solve the system completely, verifying in both original equations.

Part K — Multiple Choice and Numerical Response

17. *(Multiple Choice)* To eliminate x from $6x - 5y = 13$ and $4x + 9y = -1$, the *smallest* whole-number multipliers for the first and second equations respectively are

A. $\times 4$ and $\times 6$ B. $\times 2$ and $\times 3$ C. $\times 3$ and $\times 2$ D. $\times 9$ and $\times 5$

18. *(Multiple Choice)* The system $8x - 12y = 20$ and $-6x + 9y = k$ has infinitely many solutions when k equals

A. -20 B. -15 C. 15 D. 30

19. *(Numerical Response)* The system $7x + 4y = 47$ and $5x - 6y = 7$ has solution (x, y). The value of $25x + 2y$ is _____. *(Record your answer as a three-digit number.)*

Lesson 4 — Extra Practice

Solving Word Problems with Systems of Equations

Part A — Naming the Unknowns

*In this set the variable definitions are marked. Every solution you write must open with a line of the form "Let x be ... and let y be ...", naming the quantity **and its units**. A solution that begins with an equation and never says what the letters stand for is incomplete no matter how good the algebra that follows is. The reason is practical: it is the definition, not the arithmetic, that decides whether* 2 *belongs on the left-hand side or the right, and whether the number you end up with is an answer or nonsense.*

1. For each situation below, **do not solve anything**. Write only: **(i)** a definition of each of your two variables, in the form "Let x be ...", naming the quantity and its **units**; and **(ii)** one sentence saying which of the two variables — or which combination of them — the question is actually asking you to report.

(a) A bakery sold 74 items on Saturday, every one of them either a muffin at \$3.25 or a scone at \$4.00, taking in \$263 altogether. How many *scones* were sold?

(b) A drama club sold adult tickets at \$14 and student tickets at \$8. It sold 174 tickets and took in \$1 776. How much *more* money came from adults than from students?

(c) A cash box holds only quarters and loonies — 92 coins in all, worth \$53.75. What is the *total value of the quarters* in the box?

(d) A rectangular flower bed is edged on all four sides with 46 m of steel edging. The bed is 5 m longer than it is wide. What is its *area*?

Part B — Translation, and Where the Number Goes

Three phrases cause more lost marks in this lesson than anything else. "Twice as many A as B" puts the 2 *beside B, not beside A. "7 more than" adds; "7 times" multiplies. "The difference is* 5*" is not an equation until you say which quantity is the larger one. Test any translation you are unsure of by inventing a small pair of numbers that obeys the sentence and seeing whether your equation is true for them.*

2. For each sentence pair below, define your two variables with units and write the **two equations only**. **Do not solve any of them.**

(a) There are twice as many students as teachers on a field trip. Altogether 63 people are going.

(b) A notebook costs \$7 more than a pen. Together the two cost \$11.50.

(c) A notebook costs 7 times what a pen costs. Together the two cost \$11.20.

(d) The difference between Ravi's savings and Sofia's savings is \$5, and Ravi has the more of the two. Their savings total \$87.

(e) A jar holds nickels and dimes. There are 5 more nickels than dimes, and 61 coins in all.

(f) Three times the larger of two numbers is 8 less than five times the smaller. The two numbers add to 40.

(g) Exactly one of (b) and (c) has the pen costing more than \$2. Without solving either system, say which one, and explain how the two different phrases forced it.

3. A school council has twice as many Grade 11 representatives as Grade 10 representatives.

(a) Let e be the number of Grade 11 representatives and t the number of Grade 10 representatives. One student writes $e = 2t$; another writes $2e = t$. Decide which is correct by testing $t = 3$.

(b) The council has 18 representatives in total. Write the second equation and solve the system.

(c) Answer the question "how many representatives are in each grade?" in a sentence, and check your two numbers against *both* sentences of the problem.

Part C — Two Prices, Two Rates, Two Denominations

Three problems with the same skeleton and three different skins. In each one, one equation counts things and the other totals money — and the two must never be mixed up. Keep the units in sight: a coefficient in a counting equation is a pure number, a coefficient in a money equation carries dollars per item.

4. A plumber's first invoice lists 7 m of copper pipe and 4 elbow joints for \$108.90. A second invoice, at the same prices, lists 5 m of the same pipe and 9 of the same joints for \$110.65.

(a) Define the two variables, with units, and write the system.

(b) Solve the system, and verify your solution in **both** original equations.

(c) State both prices in a sentence with units, then find what 6 m of pipe and 5 joints would cost.

5. A community theatre charges \$18 for an adult ticket and \$11 for a youth ticket. On opening night it sold 264 tickets and took in \$3 989.

(a) Define the two variables with units and write the system. Say in a few words which of your two equations counts tickets and which totals money.

(b) Solve, and check your solution in both original equations.

(c) Now check the solution against the *words*: confirm that both counts are whole, are not negative, and add to the stated total. Then answer in a sentence with units.

6. A canteen till holds only five-dollar bills and ten-dollar bills — 68 bills worth \$475 altogether.

(a) Define the variables with units and write the system.

(b) Solve, verify in both equations, and answer in a sentence with units.

(c) The till is recounted and the supervisor reports \$478 in the same two denominations. Explain, without solving anything, why that figure cannot be right.

Part D — Numbers and Dimensions

A two-digit number is not its pair of digits. If the tens digit is t and the units digit is u, the number itself is $10t + u$, and reversing the digits produces $10u + t$. Defining t and u as the digits *is what makes those two expressions available.*

7. The two digits of a two-digit number add to 11. When the digits are reversed, the new number is 27 *less* than the original number.

(a) Define the two variables with units, write expressions for the original number and for the reversed number, and form the system.

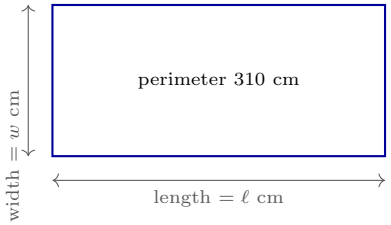
(b) Solve the system, then state the original number in a sentence. Explain why the ordered pair on its own is *not* an acceptable answer here.

(c) Check your number against both sentences of the problem, not just against your equations.

(d) Suppose the second sentence had said 28 less instead of 27 less. Solve the new system and explain what the answer means about that version of the problem.

8. A rectangular banner has a perimeter of 310 cm, and its length exceeds its width by 35 cm.

(a) Define ℓ and w with units and write the system. Take care with the order of the subtraction in the second equation.



(b) Solve the system and verify in both original equations.

(c) Answer the question "what are the dimensions of the banner?" in a sentence with units, then give its area in a second sentence with units.

(d) A classmate wrote the second equation as $\ell - w = -35$. Solve *their* system, then point to the exact words in the problem that rule their answer out.

Part E — Two-Part Plans and a Fundraiser

A charge made of a fixed part and a variable part is a linear relation in disguise: the fixed part is the y-intercept and the variable part is the slope. Comparing two such charges is a system, and its solution is the moment the two become equal — not, by itself, an answer to "which is cheaper".

9. Two phone plans are on offer. **Plan A** charges \$21 per month plus \$0.06 per minute of calling. **Plan B** charges \$9 per month plus \$0.14 per minute.

(a) Define the two variables with units and write one equation for each plan.

(b) Solve the system, and verify the solution in both original equations.

(c) Answer in a sentence with units: at what amount of calling do the two plans cost the same, and what is that monthly cost?

(d) A customer who calls about 200 minutes a month asks which plan to take. Answer with a calculation and a sentence, and explain why the solution of the system alone did not answer the question.

10. In a read-a-thon, a **Gold** sponsor pledges a flat \$25 plus \$0.40 for every book read, and a **Silver** sponsor pledges a flat \$10 plus \$0.60 for every book read. Dana read 45 books. She had 16 sponsors in total, and they pledged \$634 altogether.

(a) Show that one Gold sponsor owes \$43 and one Silver sponsor owes \$37, and say which part of each pledge is fixed and which is variable.

(b) Define the two variables with units, write the system, and solve it. Verify in both original equations.

(c) Answer in a sentence with units, then check the two counts against the words of the problem.

(d) Dana's band needs \$700. If every additional sponsor she signs is a Gold sponsor and she still reads 45 books, how many more sponsors does she need? Answer in a sentence, and justify the rounding you used.

Part F — When the Algebra Is Right and the Answer Is Not

Step 4 of the strategy is not decoration. A system can be solved perfectly and still hand back a count that is negative, a number of people that is not whole, or a pair that quietly contradicts a sentence of the story. In each case the algebra is not wrong — what is wrong is the situation, or the translation, and saying which *is part of the answer.*

11. A youth club sells hot chocolate at \$2 a cup and cookies at \$3 each, and nothing else. The Friday cash sheet records 120 items sold for \$420 in total.

(a) Define the two variables with units, write the system, and solve it.

(b) Verify that your ordered pair satisfies both equations, then state why it must be **rejected** as an answer to the question asked.

(c) Say what the rejection tells you about the cash sheet. Support it by computing the largest sum 120 items could possibly have raised.

12. A school books minivans seating 9 students each and buses seating 24 students each. It wants 14 vehicles carrying exactly 250 students, with every seat filled.

(a) Define the two variables with units, write the system, and solve it exactly — leave the coordinates as fractions.

(b) The algebra here is faultless. Explain in a sentence why the answer must still be rejected, and what that says about the booking as described.

(c) Suggest one change to the request that would make a whole-number booking possible, and give the booking it produces.

13. A stall sells kettle corn in two sizes. A **large** bag costs \$3 more than a **small** bag, and ten small bags together with six large bags cost \$78. A student defines S as the price of a small bag in dollars and L as the price of a large bag in dollars, writes

$$S = L + 3 \qquad 10S + 6L = 78,$$

solves, and reports the pair $(S, L) = (6, 3)$.

(a) Substitute $(6, 3)$ into **both** of the student's equations and show that it satisfies each one exactly.

(b) The pair passes both equations and is still wrong. Quote the sentence of the problem it contradicts, and say what that proves about checking only in the equations.

(c) Write the correct system, solve it, verify it in both equations *and* against the words, and answer in a sentence with units.

Part G — One Problem, Both Methods

14. A print shop charges a one-time setup fee plus a fixed price for each shirt. An order of 40 shirts costs \$470; an order of 65 shirts, placed on the same terms, costs \$702.50.

(a) Define the two variables with units and write the system.

(b) Solve the system by **substitution**, showing every step.

(c) Now solve the *same* system by **elimination**, showing every step. You must reach the same pair.

(d) Verify the solution in both original equations, then answer the question in a sentence with units.

(e) Give a verdict: which method suited *this* system better, and what feature of the equations decided it?

(f) Use your two values to state, in a sentence with units, what an order of 100 shirts would cost.

Part H — Error Analysis

15. A café sold 3 small coffees and 5 large coffees for \$27.10 one morning, and 6 small coffees and 2 large coffees for \$25.00 the next. A student writes:

"Let x be the number of small coffees and y the number of large coffees. Then $3x + 5y = 27.10$ and $6x + 2y = 25.00$, so $x = 2.95$ and $y = 3.65$. The café sold 2.95 small coffees and 3.65 large coffees."

(a) The student's algebra is correct and the two numbers satisfy both equations. Name the single step that went wrong, in one sentence.

(b) Write the two definitions as they should have been written, units included, and explain how the *units* of the coefficients in $3x + 5y = 27.10$ expose the error on their own.

(c) Restate the student's two numbers as a correct answer, in a sentence with units, and verify it in both original equations.

16. A stationery order of 12 binders and 5 packs of paper cost \$118.20. A second order, at the same prices, of 8 binders and 9 packs cost \$106.00. Asked for the prices, a student sets the problem up correctly, solves it correctly, and writes the complete answer as

$$(7.85,\ 4.80).$$

(a) List the three things a reader cannot tell from that answer as written.

(b) Verify that the pair does satisfy both equations, then write the answer properly as a sentence with units.

(c) Answer, in a second sentence with units, what 20 binders and 15 packs of paper would cost.

17. A charity walk raised \$4 620 from two kinds of pledge, a flat \$12 pledge and a flat \$30 pledge. There were 14 more \$12 pledges than \$30 pledges. A student defines A as the number of \$12 pledges and B as the number of \$30 pledges, writes $B = A + 14$ and $12A + 30B = 4620$, and reports $A = 100$, $B = 114$.

(a) Show that the student's pair satisfies **both** of the student's equations, and that the two counts are whole and positive.

(b) Write the relationship sentence as an equation correctly, and explain in one sentence which way round the student reversed it.

(c) Solve the corrected system, check it in the money equation *and* against the words, and answer in a sentence with units.

(d) Both pairs satisfy $12A + 30B = 4620$. Explain why checking the money alone could never have caught this error, and name the one check that does.

Part I — Stretch

18. A school store sells hoodies in three sizes: small at \$32, medium at \$38 and large at \$45. On one day it sold 54 hoodies for \$2 124, and the number of large hoodies sold was exactly **twice** the number of small hoodies sold.

(a) Define **three** variables with units and write the three facts as three equations.

(b) Use the stated relationship to eliminate one variable, reducing the problem to a system of two equations in two unknowns. Solve it.

(c) Recover the third quantity, check all **three** conditions, and answer in a sentence with units.

(d) Explain why the relationship "twice as many large as small" had to be written $\ell = 2s$ and not $2\ell = s$, using the numbers you found as the test.

19. Two workers spread mulch. Working on their own, Amir covers 11 m^2 per hour and Bo covers 8 m^2 per hour. On Saturday the two of them logged 15 hours of work between them and covered 141 m^2.

(a) Define the two variables with units. Build one equation from the **total time** and the other from the two **rates**, and say which is which.

(b) Solve the system and verify in both original equations.

(c) Answer in a sentence with units, and check both hour counts against the words of the problem.

(d) The supervisor's log for the following Saturday claims the same two workers logged 15 hours between them and covered 170 m^2. Solve that system, and use the answer — together with the largest area 15 hours could possibly cover — to explain what must be wrong with the log.

Lesson 5 — Extra Practice

Mixture and Motion: More Applications of Linear Systems

Part A — The Mixture Table

*Every mixture problem carries the same two facts, and a three-column table pulls both of them out of the words. One column holds the **amount** of each ingredient, one holds its **concentration** (or its **unit value**, in dollars per kilogram), and the third is the product of the first two — the **content** that ingredient contributes. The two equations are then read straight off the table: the amount column adds to the total amount, and the content column adds to the total content. Those two are always the equations. Nothing else is.*

1. A metal shop stocks an alloy that is 35% copper and an alloy that is 80% copper. It needs 180 kg of an alloy that is 50% copper. Let x be the kilograms of the 35% alloy and y the kilograms of the 80% alloy.

(a) Complete the table.

Ingredient	**Amount (kg)**	**Concentration**	**Copper (kg)**
35% alloy			
80% alloy			
Mixture			

(b) Read the two equations off the table, and name which column each one comes from.

(c) Solve the system. Show the step that clears the decimals to integers.

(d) Check your answer in **both** equations, then answer in a sentence with units.

2. A supplier blends a fertilizer that is 22% nitrogen with one that is 10% nitrogen to produce 300 kg of a blend that is 14% nitrogen. **Do not solve this system.**

(a) Complete the table, using m for the kilograms of the 22% fertilizer and n for the kilograms of the 10% fertilizer.

Ingredient	**Amount (kg)**	**Concentration**	**Nitrogen (kg)**
22% fertilizer			
10% fertilizer			
Blend			

(b) Write the two equations, and state in words what each one counts.

(c) The number 14 is not halfway between 10 and 22. Without solving, say which fertilizer there is more of, and why.

(d) A classmate writes $22m + 10n = 14$ as the second equation. Explain what is wrong with the right-hand side.

3. The table works just as well when the middle column is a *price* instead of a percent. A shop blends cashews worth \$26/kg with peanuts worth \$8/kg to make 45 kg of a mixture worth \$14/kg.

(a) Build the table yourself — three rows, three columns — and write the two equations from it.

(b) Solve the system and check both equations.

(c) Answer in a sentence with units, and state the total value of the blend in dollars.

(d) In question 1 the third column was measured in kilograms; here it is measured in dollars. Explain why the *same* table still works.

Part B — When One Ingredient Is Pure

A pure ingredient is not a special case; it is the ordinary case with a concentration of 100% *or* 0%*. Pure concentrate is all content: its middle-column entry is* 1*. Pure water carries volume but no content at all: its middle-column entry is* 0*, so it contributes nothing to the second equation — but it still appears in the first.*

4. A mechanic has a premixed coolant that is 30% antifreeze and a drum of pure (100%) antifreeze concentrate. He needs 21 L of coolant that is 50% antifreeze.

(a) Define both variables with units, give the middle-column entry for each ingredient, and write the two equations.

(b) Solve the system and check in both equations.

(c) Answer in a sentence with units, and check the answer against the context as well as the algebra.

5. A greenhouse has a fertilizer solution that is 24% concentrate. It must be diluted with **pure water** to produce 40 L of a solution that is 15% concentrate.

(a) Write the two equations. State the concentration you used for the water and justify it.

(b) Solve the system and check in both equations.

(c) Answer in a sentence with units.

(d) A student objects that "water has no equation, so there is only one unknown." Explain why water does belong in the first equation, and what would go wrong with the total volume if it were left out.

Part C — Is the Answer Even Possible?

A mixture cannot be stronger than its strongest ingredient or weaker than its weakest one. The final concentration is a weighted average, so it must land between *the two, and closer to whichever ingredient there is more of. When a target lies outside that interval, the algebra does not fail — it returns a negative amount, which is the algebra's way of saying the recipe is impossible.*

6. A chemistry technician has a 20% solution and a 45% solution of the same salt.

(a) Write and solve a system for 50 L of a 30% solution. Check in both equations and answer in a sentence with units.

(b) Now write and solve the system for 50 L of a 60% solution. Report exactly what the algebra gives you.

(c) Interpret the answer to (b) in the context. What is the technician being asked to do with the 20% solution?

(d) State, with a reason, the complete set of concentrations that *can* be made from these two solutions alone. Which two targets sit exactly at the ends of that set, and how would each of them be made?

7. A fish hatchery has brine at 12% and brine at 30%. For each order below, decide **before calculating** whether it is possible, and say why.

(a) 600 L at 24%
(b) 600 L at 9%
(c) 600 L at 30%

(d) Solve the one order in (a)–(c) that needs two genuine ingredients. Check in both equations and answer in a sentence with units.

(e) Your answer to (d) uses more of one brine than the other. Show that this agrees with where 24% sits between 12% and 30%.

Part D — Moving With and Against a Current

A current or a wind does not change how hard the engine works; it changes the speed over the ground. With b the speed in still water or still air and c the speed of the current or wind, the ground speed is $b + c$ with it and $b - c$ against it. Those two expressions, each multiplied by its own time, give the two distances — and there is the system.

8. A freight barge travels 90 km downstream in 3 hours and the same 90 km back upstream in 5 hours.

(a) Find the ground speed on each leg, then write a system in b and c.

(b) Solve the system and check in both equations.

(c) State both speeds in a sentence with units, and say how long the 90 km would take on still water.

9. A cargo plane flies 1500 km with a steady tailwind in 2.5 hours. It returns the same 1500 km against that same wind in 3 hours.

(a) Let p be the plane's speed in still air and w the wind speed, both in km/h. Write the two equations using distance = speed × time.

(b) Solve the system and check in both equations.

(c) Answer in a sentence with units. Then explain why the round trip takes longer than it would with no wind at all, even though the wind helps on one leg.

10. In each part below the numbers arrive in mismatched units. Reconcile the units **first**, then build and solve the system.

(a) A rowing shell covers 1800 m downstream in 6 minutes and 1800 m upstream in 9 minutes. Find the shell's speed in still water and the speed of the current, first in m/s as exact fractions and then in km/h.

(b) A technician has an 8% hydrogen peroxide solution and a 35% solution. She needs 900 mL of a 20% solution, and has only 1.2 L of the 8% solution on the shelf. Find the volume of each solution required, and state whether the order can be filled.

(c) In part (a), what would have happened if 6 and 9 had been used as though they were hours? Give the number the current would have come out to, and say why it is obviously wrong.

Part E — Two-Leg Journeys

When a trip is made of two legs, the two equations come from two different totals: the two distances add to the total distance, and the two driving times add to the total driving time. Driving time is not elapsed time — every stop has to come out of the total before the time equation is written.

11. A driver covers 250 km between two towns, part of it on gravel at 30 km/h and the rest on pavement at 80 km/h. The trip takes 3 hours 50 minutes in total, including one 20-minute stop.

(a) Find the total *driving* time in hours, then write a system in g (hours on gravel) and p (hours on pavement).

(b) Solve the system and check in both equations.

(c) State the distance travelled on each surface in a sentence with units.

12. A cyclist leaves a trailhead at 7:00 a.m. riding at a steady 18 km/h. A support car leaves the same trailhead along the same route at 8:30 a.m. travelling at 54 km/h.

(a) Let t be the hours since 7:00 a.m. and d the distance from the trailhead in km. Write one equation for each vehicle.

(b) Solve the system and check in both equations.

(c) State the time of day the car catches the cyclist and how far along the route it happens, in a sentence with units.

(d) Explain why the car's equation uses $t - 1.5$ rather than t, and what the point where the two lines cross means on a distance–time graph.

Part F — Error Analysis

13. Asked to mix a 15% and a 40% solution to make 200 mL of a 25% solution, a student wrote

$$x + y = 200 \qquad 0.15x + 40y = 25(200).$$

(a) Name the error in one sentence, and say what the three rates would have to mean for the student's equation to be true.

(b) Write the content equation twice — once entirely in decimals, once entirely in whole-number percents — and show the two are the same equation.

(c) Solve the system correctly, check in both equations, and answer in a sentence with units.

14. A jet boat covers 60 km downstream in 1.5 hours and 60 km upstream in 2.5 hours. A student wrote

$$1.5(b + c) = 60 \qquad 2.5(b + c) = 60.$$

(a) Solve the student's system as written and classify it. What does the classification say about the two lines?

(b) Explain, in terms of the boat and the river, why the second equation must use $b - c$.

(c) Fix the system, solve it, check in both equations, and answer in a sentence with units.

15. A shop blends tea worth \$30/kg with tea worth \$18/kg to make 50 kg of a blend worth \$22.80/kg. A student reports "32 kg of the \$30 tea and 22 kg of the \$18 tea."

(a) Test the student's answer in **both** equations. Which one fails first, and by how much?

(b) Explain why the amount equation is the faster of the two checks, and why it should be run before the value equation every time.

(c) Solve the system correctly, check in both equations, and answer in a sentence with units.

Part G — Stretch

16. A trail mix is made from almonds worth \$22/kg, raisins worth \$7/kg, and pretzels worth \$4/kg. The finished mix weighs 40 kg and is worth \$12.10/kg. The mix contains **equal masses** of raisins and pretzels.

(a) There are three ingredients but only two unknowns. Explain how the stated relationship does that, and define your two variables with units.

(b) Write the amount equation and the value equation, then solve the system.

(c) Check all three masses and the total value, and state the recipe in a sentence with units.

17. Two boats work the same stretch of river, which has one steady current. A motorboat travels 72 km downstream in 3 hours. A second boat, whose speed in still water is 4 km/h *less* than the motorboat's, travels 42 km upstream in 3 hours.

(a) Let b be the motorboat's still-water speed and c the current, both in km/h. Write the second boat's still-water speed in terms of b, then write the two equations.

(b) Solve the system and check in both equations.

(c) State the current and both still-water speeds in a sentence with units, then find how long the second boat would take to travel the same 42 km downstream.

Part H — Multiple Choice and Numerical Response

18. *(Multiple Choice)* A juice drink is 40% juice. Pure water is added to 15 L of it to bring it down to 25% juice. The number of litres of water added is

A. 6 B. 9 C. 10 D. 24

19. *(Multiple Choice)* A canoeist paddles at a steady speed in still water. On a river she covers a stretch in 40 minutes going downstream and the same stretch in 60 minutes going upstream. The ratio of her still-water speed to the speed of the current is

A. 2 : 1 B. 3 : 2 C. 5 : 1 D. 6 : 1

20. *(Numerical Response)* Beans worth \$16/kg are blended with beans worth \$25/kg to make 36 kg of a blend worth \$19/kg. The number of kilograms of the \$25/kg beans in the blend is _____. *(Record your answer as a three-digit number.)*

Lesson 6 — Extra Practice

Systems of Linear Equations Unit Review

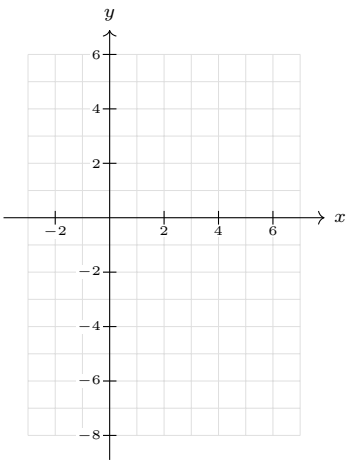
Part A — One System, Every Method

Three methods, one system, one answer. A method can be faster or slower, cleaner or messier, but it can never change what the answer is *— if two methods disagree, one of them contains an arithmetic error, not a difference of opinion. Question 1 is the whole unit in a single problem: graph it, substitute it, eliminate it, then judge which of the three you would actually have chosen.*

1. Consider the system

$$3x + 2y = 5 \qquad 2x - 3y = 12$$

(a) **By graphing.** Rewrite both equations in slope y-intercept form, graph both lines on the grid, and state the solution you read off.



(b) **By substitution.** Isolate one variable, substitute into the other equation, and solve. Carry every fraction exactly — no decimals at any stage.

(c) **By elimination.** State the lowest common multiple you are using and the two multipliers, then eliminate and solve.

(d) Set your three answers side by side. Do all three agree? Verify the common answer in **both original** equations.

(e) State the number of solutions this system has and say why, quoting both slopes. Then state what the *product* of the two slopes tells you about the pair of lines.

(f) Which of the three methods suited *this* system best, and why? Then describe one change to the system that would have made you name a different method.

Part B — Choosing a Method

Choosing well is a skill of its own, and it is exercised before *any algebra happens. A variable already isolated, or a coefficient of* 1*, points to substitution. Coefficients that match, or that scale easily to match, point to elimination. And when the question asks only* roughly *where two lines meet, a graph answers it faster than either.*

2. For each system, state which method — graphing, substitution, or elimination — you would use, and give one sentence of justification that refers to the actual coefficients. **Do not solve these here.**

(a) $y = 5 - 2x$
$3x + 4y = 10$

(b) $5x - 6y = 39$
$5x + 2y = 7$

(c) $x - 7y = 9$
$4x + 3y = 5$

(d) $7x + 4y = 24$
$5x - 6y = 26$

3. Now solve all four systems of question 2 by the method you named, and verify each solution in **both** of its original equations.

(a) system 2(a)

(b) system 2(b)

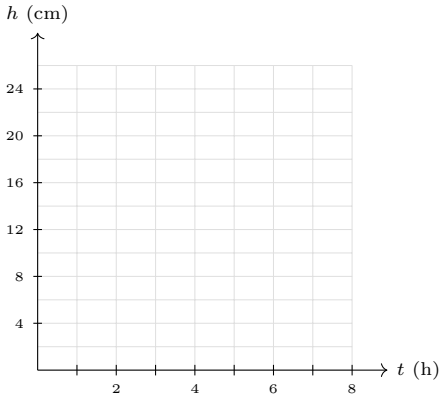
(c) system 2(c)

(d) system 2(d)

(e) All four solutions turned out to be lattice points, yet graphing would still have been a poor choice for system 2(b). Write both of its equations in slope y-intercept form and explain why.

4. Two candles are lit at the same moment. The taller one starts at 25 cm and burns down 4 cm every hour; the shorter one starts at 18 cm and burns down 2.5 cm every hour. Let t be the time in hours since they were lit and h the height of a candle in centimetres. A stage manager does not need an exact answer — she only needs to know, *roughly*, when the two candles will look the same height.

(a) Write an equation for each candle and graph both on the grid.



(b) From your graph, estimate when the candles are the same height, to the nearest half hour, and estimate that common height to the nearest centimetre. Answer in a sentence with units.

(c) Explain why a graph is an entirely reasonable tool for the stage manager's question, even though Lesson 1 warned that graphs give only estimates.

(d) Now find the exact answer algebraically, as fractions in lowest terms, and verify it in both of your equations. State how far your graphical estimate was from the exact time.

Part C — Classifying and Solving for a Parameter

A system carrying an unknown coefficient can be classified without ever comparing slopes. Eliminate a variable as usual and watch what survives: an equation of the form (number) × (variable) = (number). If the coefficient on the left can be driven to zero, the system is one of the two special cases — which one depends entirely on whether the right-hand side went to zero with it.

5. In the system $3x - 4y = 7$ $kx + 8y = 5$, k is a constant.

(a) Eliminate y and show that the system reduces to $(k + 6)x = 19$.

(b) Determine the value of k for which the system has **no solution**. Justify it twice: once from the surviving equation in (a), and once by comparing slopes.

(c) Show that **no** value of k gives infinitely many solutions, and name the condition that fails.

(d) Determine the value of k for which the x-coordinate of the solution is 1. State the complete solution and verify it in both equations.

6. In the system $4x + 6y = 10 \qquad 10x + ky = c,$ both k and c are constants.

(a) Eliminate x and show that the system reduces to $(30 - 2k)\,y = 50 - 2c$.

(b) Determine the values of k and c for which the system has **infinitely many solutions**, and state the multiplier that links the two equations.

(c) State the condition on k and c that gives **no solution**, and the condition that gives **one solution**. Use the equation from (a), not the slopes.

(d) A classmate says "if $k = 15$ then the system has no solution." Say exactly when the classmate is right and exactly when the classmate is wrong.

Part D — Modelling a Situation

Nothing here tells you which method to use — that decision is yours, and it is part of the question. Define each variable with its unit*, write one equation for each condition in the words, solve by whichever method the numbers deserve, check in both equations, and finish with a sentence a reader could act on.*

7. At Ravenscliff High the final mark in a course is built from two pieces: the coursework mark counts for 40% and the final exam mark counts for 60%. Amara's exam mark was 14 percentage points higher than her coursework mark, and her final mark came out at 81.4.

(a) Define two variables with their units and write a system of two equations.

(b) Name the method you will use and say why, then solve. Clear the decimals first if that helps.

(c) Verify your solution in both original equations, then answer the question in a sentence with units.

(d) With the same exam mark, what coursework mark would Amara have needed for a final mark of exactly 85? Answer in a sentence.

(e) Without solving anything further, explain how the two equations alone guarantee that this situation has exactly one answer rather than many.

8. A steel spring hangs from a hook in a physics lab. With a 3 kg mass attached its length is 26 cm; with a 7 kg mass attached its length is 38 cm. Within this range the length L, in centimetres, is modelled by $L = a + bm$, where m is the attached mass in kilograms and a and b are constants.

(a) Substitute each measurement into $L = a + bm$ to form a system of two equations in a and b.

(b) Solve the system, choosing your method and saying why, and verify a and b in both equations.

(c) State what a and b each mean for this spring, in a sentence with units, and predict the length with a 5 kg mass.

(d) A second spring on the same hook is modelled by $L = 23 + 2m$. Determine the mass at which the two springs have the same length, and state that common length. Answer in a sentence with units.

(e) Classify the system formed by the two spring models, and explain what a "no solution" answer would have meant about the two springs physically.

Part E — Reasoning and Error Analysis

*Every sentence below sounds plausible, and every piece of work below was done by a student who was concentrating. A claim is settled by a counter-example, not by how reasonable it sounds; a piece of work is settled by substituting the answer back into **both** original equations.*

9. For each statement, state whether it is **always** true, **sometimes** true, or **never** true. Justify each answer, and wherever the statement can fail, give a specific counter-example — a system, a pair of lines, or a point.

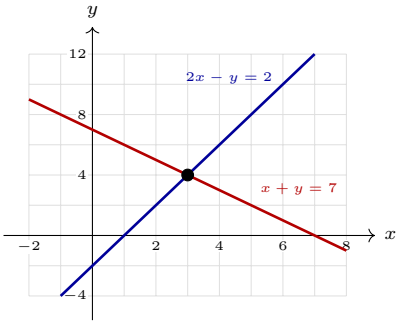
(a) "A system made of two *different* linear equations has exactly one solution."

(b) "If elimination reduces a system to $0 = 0$, then the solution is $(0, 0)$."

(c) "Graphing is the safest method for solving a system, because you can see the answer on the page."

(d) "Two lines with different y-intercepts must cross somewhere."

10. The system $2x - y = 2$ and $x + y = 7$ is graphed below, with the point of intersection marked. **On this grid one square is 1 unit wide and 2 units high.** A student reports the solution as $(3, 2)$.



(a) Substitute $(3, 2)$ into both equations and show that it fails both. Then name, in one sentence, exactly what the student did wrong when reading the graph.

(b) Read the marked point correctly, state the solution as an ordered pair, and verify it in both original equations.

(c) Solve the system algebraically, by either method, and confirm that it agrees with the point you read in (b). Then state the one thing you must check on any grid before reading a coordinate from it.

11. A bicycle shop charges a flat \$45 for a tune-up and \$20 for a safety check. Last week the shop performed 38 of these services altogether and collected \$1360 for them. Asked to find how many of each were performed, a student writes the system

$$45x + 20y = 38 \qquad x + y = 1360$$

where x is the number of tune-ups and y is the number of safety checks.

(a) Name the student's error in one sentence, and write the system as it should have been written.

(b) Solve your corrected system by whichever method you judge best, and verify the solution in both of your equations.

(c) Answer the shop's question in a sentence with units, and check the numbers against the *words* of the problem as well as against the equations.

(d) Solve the student's system as written. Both coordinates come out impossible for this situation — say what is wrong with each of them, and explain why a check against the words catches an error that a check against the equations never would.

Part F — Challenge

Both questions below carry an unknown constant all the way through the solution, so the answer is not a point but a formula *for a point. Once you have that, extra conditions can be imposed on it — a third line to sit on, or a demand that a coordinate come out whole.*

12. In the system $2x + y = k \qquad 3x - 2y = 12,$ k is a constant.

(a) Solve the system in terms of k, writing each coordinate as a single fraction with denominator 7.

(b) Determine the value of k for which the solution also lies on the line $y = x - 5$, and state the solution point.

(c) Verify that your point from (b) satisfies all *three* equations.

(d) Explain why there is no value of k at all for which the solution lies on the line $3x - 2y = 0$.

13. In the system $kx + 3y = 12$ $2x - y = 4$, k is an **integer**.

(a) Show that the system reduces to $(k+6)\,x = 24$, and state the one value of k for which the system has no solution.

(b) Determine **every** integer k for which the x-coordinate of the solution is a whole number. List them in order.

(c) Take the largest k on your list, solve the system completely, and verify the solution in both original equations.

(d) Explain why the list in (b) has to stop — what happens to x for every k larger than that largest value? Explain also why $x = 0$ can never occur.

Part G — Loose Ends From Lessons 1 to 5

The questions above did not need every technique the unit taught. These do: an equation that must be tidied before a method can start, a journey, a mixture, a number puzzle, a shape, an investment, and the short-answer formats the exam uses.

14. Neither system below will accept any method until it has been tidied up — one is buried in fractions, the other in brackets. Clear it, choose a method, solve, and verify your solution in the **original** equations as printed.

(a)
$$\frac{2x - y}{5} + \frac{x + 3y}{2} = 4$$
$$\frac{x}{4} - \frac{y}{3} = \frac{5}{12}$$

(b)
$$2(x - 3y) + 5 = 3(x + y) - 4$$
$$4(2x + y) - 7 = 5(x - y) + 20$$

15. An airport moving walkway runs 120 m from one end to the other. Walking *with* the walkway, Nadia covers its full length in 40 s. Walking *against* it at the same walking pace, she covers the same 120 m in 120 s.

(a) Let w be Nadia's walking speed and c the speed of the walkway, both in m/s. Using speed = distance ÷ time, write a system of two equations.

(b) Solve the system by elimination and verify both values in both original equations.

(c) State both speeds in a sentence with units.

(d) How long would the walkway take to carry a passenger who just stands still on it, and how long would Nadia take to walk 120 m on ordinary floor? Answer both in a sentence with units.

16. A dairy blends cream that is 32% butterfat with skim milk that is 2% butterfat to produce 60 L of a product that is 10% butterfat.

(a) Define two variables with their units and write a system — one equation for total volume, one for total butterfat.

(b) Clear the decimals, solve by elimination, and verify in both original equations.

(c) State how many litres of each ingredient are needed, in a sentence with units.

(d) Explain why no blend of these two ingredients could ever be 35% butterfat, no matter what volumes are used.

17. In a certain fraction the numerator is 5 less than the denominator. When 3 is added to *both* the numerator and the denominator, the new fraction is exactly $\frac{4}{5}$.

(a) Let n be the numerator and d the denominator. Write a system of two equations, clearing the fraction in the second one.

(b) Solve the system and state the original fraction. Check it against the *words*, not just against the equations.

(c) A classmate says any fraction equivalent to your answer would do just as well. Double both your numerator and your denominator, test that fraction against *both* sentences, and explain what goes wrong.

18. Two more situations, each needing a system of two equations. For each: define the variables with units, write the system, solve by the method you judge best, verify in both equations, and answer in a sentence with units.

(a) An isosceles triangle has a perimeter of 74 cm. Each of its two equal sides is 7 cm more than twice the length of the base. Determine the base and each equal side.

(b) Mei invested \$7500 in two places: part in a savings bond paying 3% per year, and the rest in a term deposit paying 4.5% per year. After one year the two together had earned \$285 in interest. Determine how much went into each.

19. *(Multiple Choice)* One system of two linear equations is handed to three students. The first solves it by graphing and reports $(2, 5)$; the second solves it by substitution and reports $(2.5,\ 4.75)$; the third solves it by elimination and reports $\left(\frac{5}{2}, \frac{19}{4}\right)$. Which statement is correct?

A. The grapher must be right, since a graph shows the actual point of intersection.
B. All three are right, because each used a different method.
C. The second and third students agree exactly, and the first has misread a point that no one-unit grid can display.
D. The system has infinitely many solutions, since three different answers were obtained.

20. *(Numerical Response)* The system $5x + 2y = 38$ and $3x - 4y = 2$ has solution (x, y). The value of $20x + 6y$ is _____. *(Record your answer as a three-digit number.)*